SOCIAL MOOD MATCHER

AI-Powered Social Media Caption & Hashtag Generator

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1. Introduction

Social media platforms require engaging captions and relevant hashtags to improve visibility and audience engagement. However, creating captions manually for every post is time-consuming and often inconsistent.

The Social Mood Matcher project is an AI-powered system that automatically generates captions and hashtags based on the mood and context of an uploaded image.

The system uses advanced AI models:

- BLIP for image captioning
- DistilBERT for sentiment analysis
- Gemini 2.5 Flash (optional) for enhanced caption generation

The generated content is automatically optimized for different social media platforms such as Twitter/X, Instagram, and Facebook.

2. System Workflow (Output Step Process)

The Social Mood Matcher system follows a step-by-step workflow to process images and generate captions.

<u>Step 1: Application Interface</u>

When the user launches the application, the Streamlit user interface is displayed.

Users can upload an image and select the desired caption style and platform.

Features available in the interface:

- Image upload option
- Caption style selection
- Platform selection
- Hashtag count adjustment
- Gemini AI toggle option

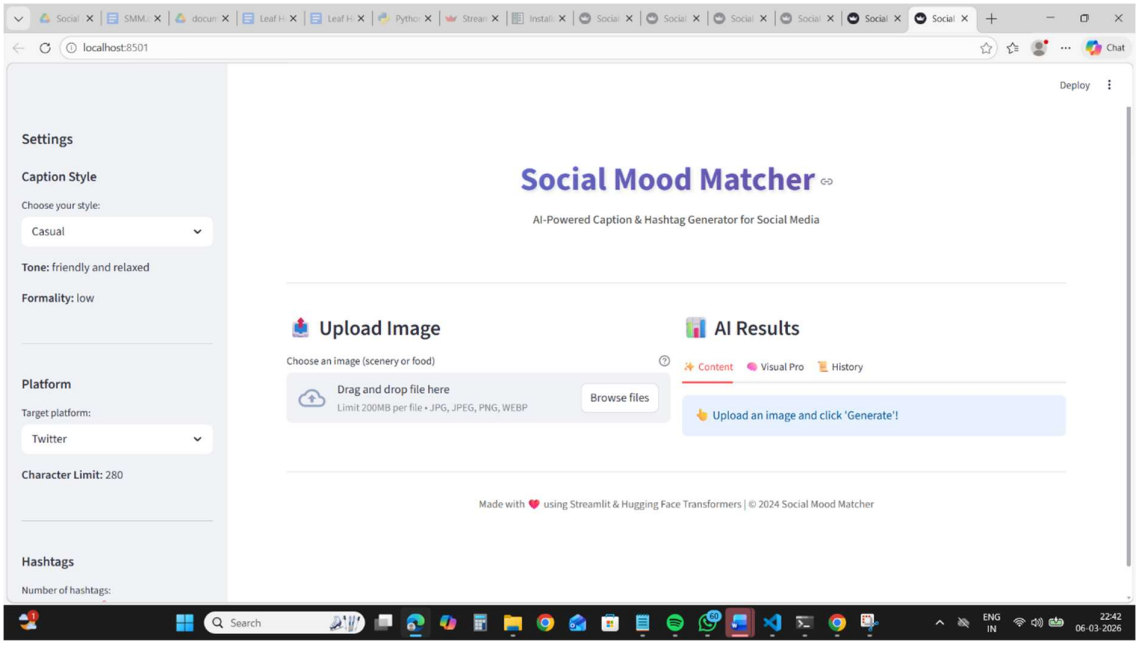

Fig 1.1 Main Application Interface

<u>Step 2: Image Upload and Preview</u>

The user uploads an image (JPG, PNG, or WEBP format).

The system validates the image and displays a preview before processing.

Functions performed in this step:

- Image validation
- Image resizing
- Thumbnail preview generation
- File size verification (Max: 10MB)

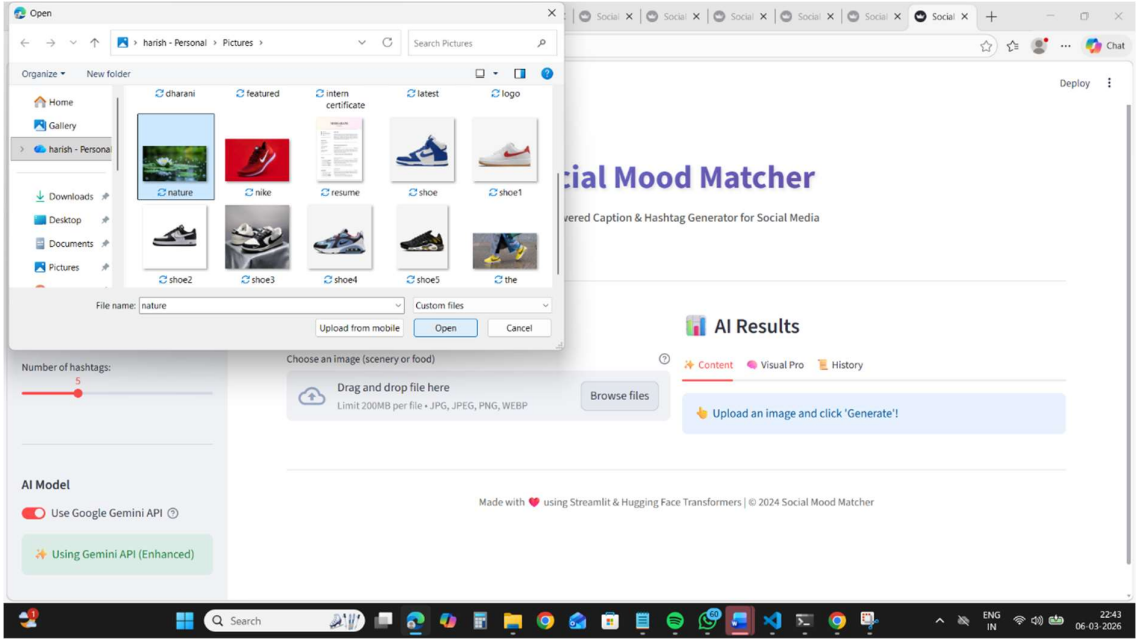

Fig 1.2 Image Upload & Preview

<u>Step 3: Image Analysis using AI Models</u>

After the image is uploaded, the system analyzes the image using BLIP to generate a visual description.

Example output:

"A water lily in a pond"

This caption represents what the AI understands from the image.

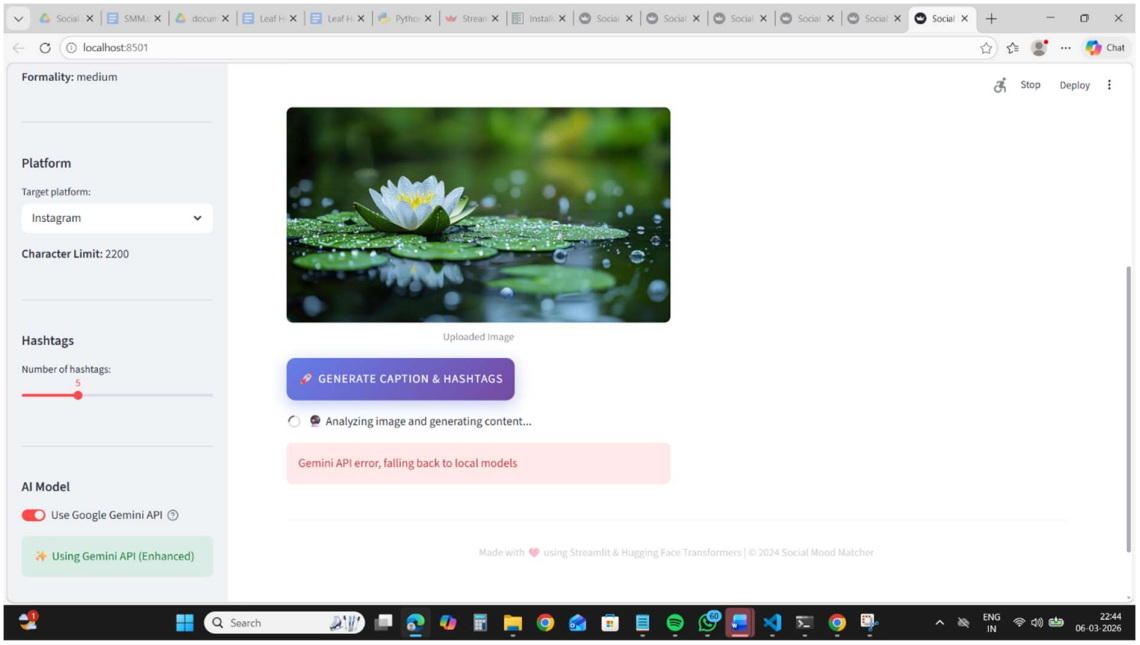

Fig 1.3 Image Caption Detection

<u>Step 4: Sentiment Detection</u>

The generated caption is passed to the DistilBERT sentiment model to determine the emotional tone of the image.

Possible detected moods include:

Happy

Cozy

Calm

Aesthetic

Romantic

Energetic

Nostalgic

Luxury

Peaceful

Adventurous

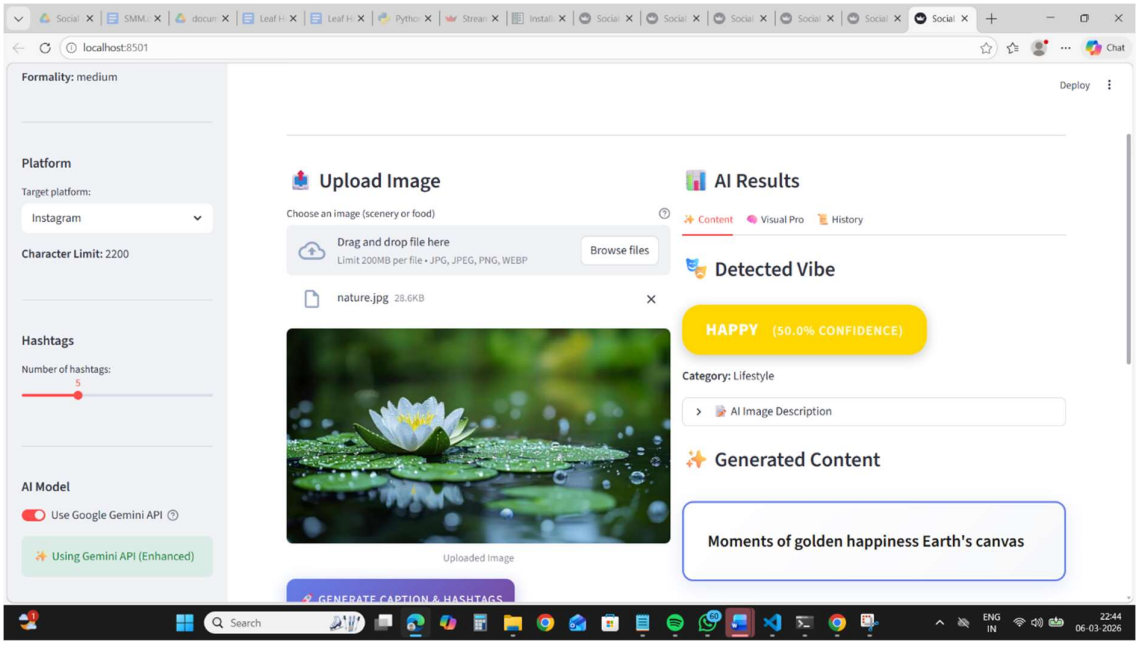

Fig 1.4 Sentiment Detection Output

<u>Step 5: Caption Generation</u>

Based on the detected sentiment, the system generates captions according to the selected caption style.

Available caption styles:

Style Description

Casual Friendly and relaxed

Aesthetic Artistic and poetic

Professional Formal and informative

Playful Fun and energetic

Example generated captions:

Casual:

Nothing beats a warm bowl of homemade soup on a chilly evening.

Aesthetic:

Comfort served in every spoonful.

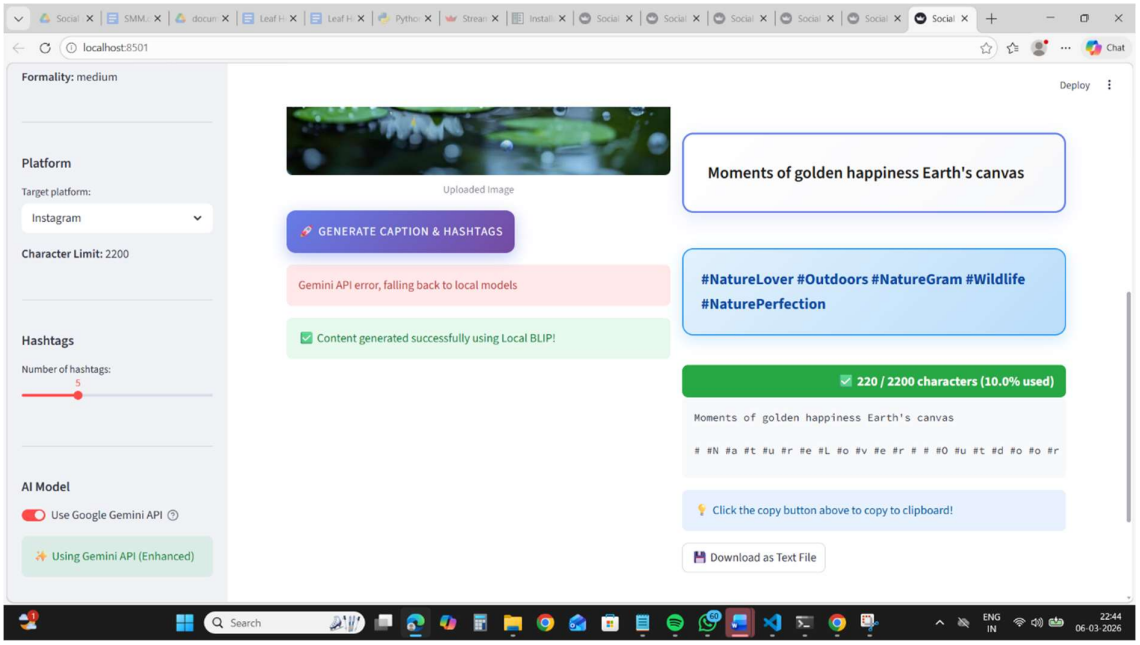

Fig 1.5 Generated Caption

<u>Step 6: Hashtag Generation</u>

The system generates trending hashtags based on:

- Image category
- Sentiment
- Popular social media trends
- Example hashtags:
- #FoodPhotography
- #ComfortFood
- #CozyVibes
- #Homemade
- #Foodie

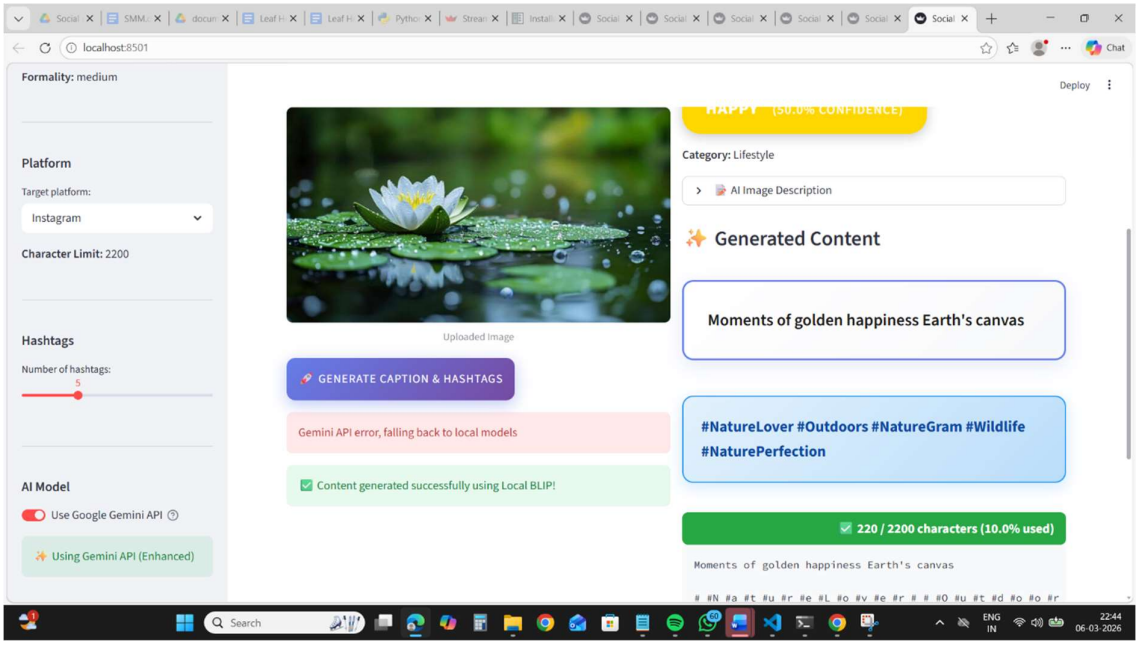

Fig 1.6 Hashtag Output

<u>Step 7: Platform Character Limiting</u>

Each social media platform has different character limits.

Platform	Character Limit
Twitter/X	280
Instagram	2200
Facebook	63206

The system automatically ensures that captions and hashtags remain within the allowed limits.

If necessary, hashtags are truncated while keeping the caption intact.

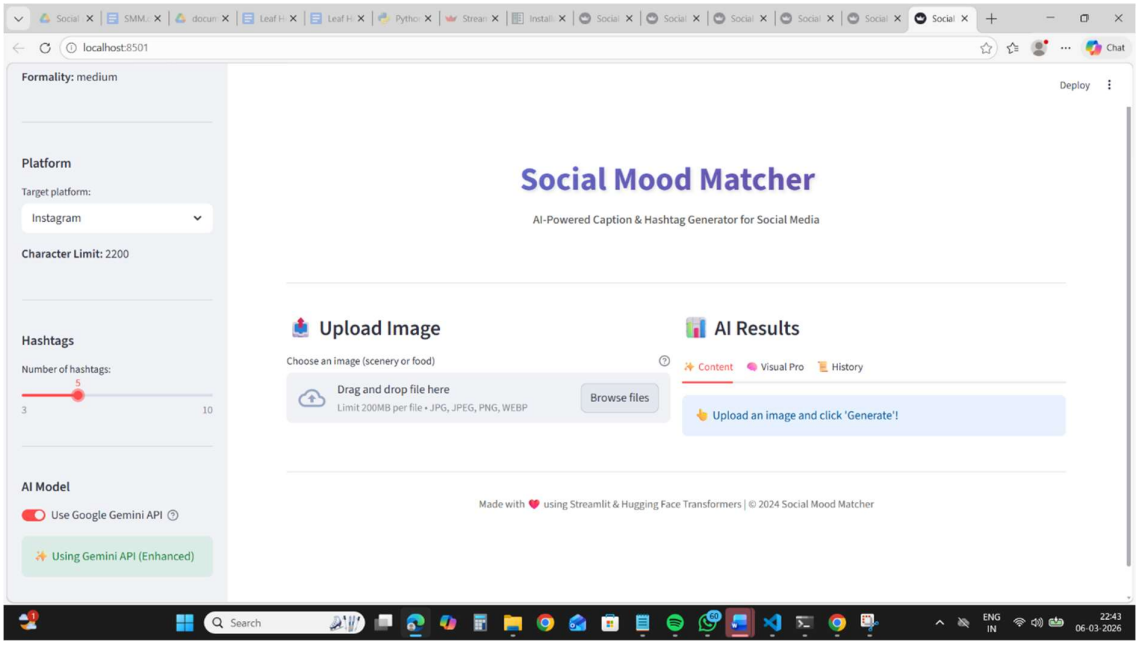

Fig 1.7 Character Limit Display

<u>Step 8: Final Output Display</u>

The final output includes:

- Generated caption
- Suggested hashtags
- Character count
- Sentiment result

The user can copy the content or download it.

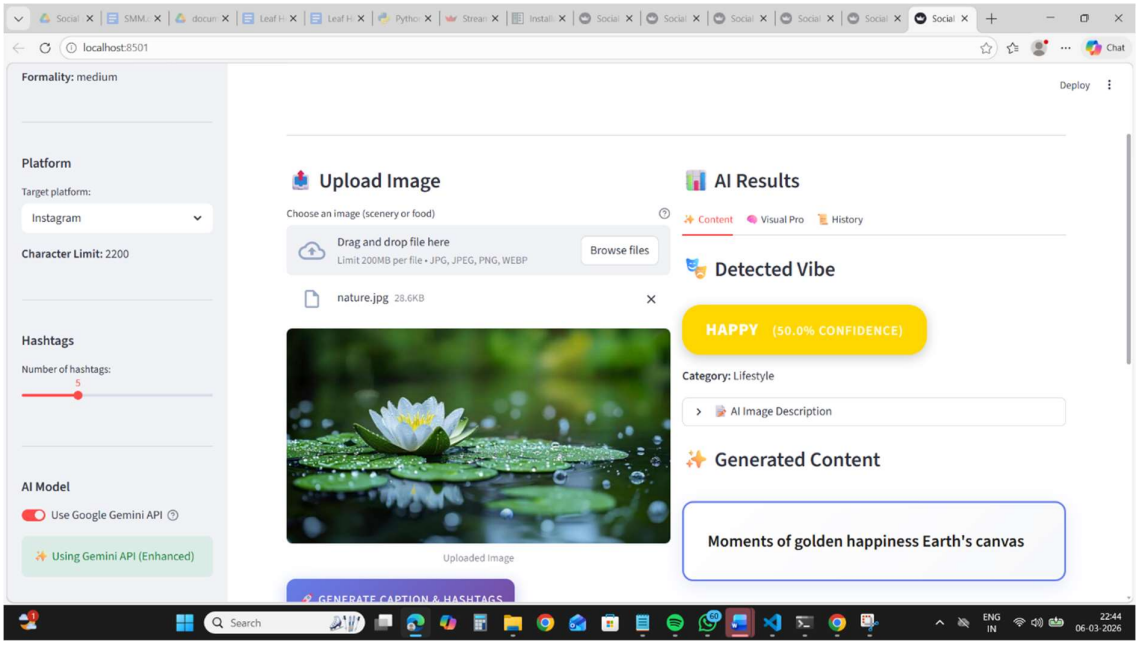

Fig 1.8 Final Output Screen

<u>Step 9: Export and Download</u>

The generated content can be exported in different formats:

- Text file
- Word document
- PDF file
- This allows users to easily reuse captions across multiple platforms.

3. System Architecture

The Social Mood Matcher system follows a modular architecture.

Components include:

Component	Function
Streamlit UI	User interaction and display
Image Sentiment Detector	Image understanding and mood detection
Caption Generator	Creates captions based on sentiment
Hashtag Engine	Suggests trending hashtags
Character Limiter	Ensures platform compatibility
Gemini AI Layer	Optional advanced caption generation

4. Technologies Used

Technology	Purpose
Python	Backend development
Streamlit	Frontend UI
BLIP	Image captioning
DistilBERT	Sentiment analysis
Gemini API	Advanced AI generation
Transformers	AI model integration
Pillow	Image processing
Python-DOCX	Word export
ReportLab	PDF generation

5. Conclusion

Social Mood Matcher simplifies social media content creation by using AI to analyze images and generate captions automatically.The system combines image understanding, sentiment analysis, and generative AI to produce engaging social media content tailored to the mood of the image.By automating caption creation and hashtag selection, the platform helps content creators save time while improving engagement on social media platforms.